# Ryan E. Mills, PhD

**Professor and Associate Chair, Department of Computational Medicine and Bioinformatics**
**Professor, Department of Human Genetics**
**1301 Catherine St, Room 5B390**
**Ann Arbor, MI 48109**
**734-647-9628 - remills@umich.edu**


## Education and Training

### Education

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|---|---|
| 08/1996-05/2000 | AB, Wabash College, Crawfordsville, IN |
| 08/2000-08/2003 | MS, Georgia Institute of Technology, Atlanta, GA |
| 09/2003-05/2006 | PhD, Georgia Institute of Technology, Atlanta, GA |

### Postdoctoral Training

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| 01/2006-10/2008 | Postdoctoral Fellow, Genetics/Genomics, Emory University, Biochemistry, Atlanta, GA, Dr. Scott Devine |

## Work Experience

### Academic Appointment

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|---|---|
| 01/2012-09/2018 | Assistant Professor, Computational Medicine and Bioinformatics, University of Michigan, Ann Arbor, MI |
| 01/2012-09/2018 | Assistant Professor, Human Genetics, University of Michigan, Ann Arbor, MI |
| 09/2018-08/2024 | Associate Professor, Human Genetics, University of Michigan, Ann Arbor, MI |
| 09/2018-08/2024 | Associate Professor, Computational Medicine and Bioinformatics, University of Michigan, Ann Arbor, MI, (Tenured) |
| 09/2024-Present | Professor, Computational Medicine and Bioinformatics, University of Michigan, Ann Arbor, MI, (Tenured) |
| 09/2024-Present | Professor, Human Genetics, University of Michigan, Ann Arbor, MI |

### Administrative Appointment

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|---|---|
| 01/2009-12/2011 | Team Leader, Bioinformatics and Medical Diagnostics Team, Molecular Genetic Research Unit, Brigham and Women's Hospital, Boston, MA |
| 04/2022-Present | Program Director of DCMB Computing Infrastructure, Computational Medicine and Bioinformatics, University of Michigan, Ann Arbor, MI |
| 02/2025-Present | Associate Chair for Research, Computational Medicine and Bioinformatics, University of Michigan, Ann Arbor, MI |

### Research Position

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|---|---|
| 11/2008-12/2011 | Research Associate, Pathology, Brigham & Women's Hospital, Harvard Medical School, Boston, MA |

## Research Interests

- The primary purpose of sequencing genomes is to identify the underlying genetic variation both

between and within individuals and to explore what role those changes have on human phenotypes. Our research laboratory develops and implements methods to precisely identify and resolve different types of genomic variation. Our goal is to integrate this information with other forms of biologically and medically relevant data to improve our overall understanding of human health and disease.

## Grants

**Current Grants**

R01DE034740:*Understanding Clonal Enrichment and Functional Impact of HPV Integration Events in OPSCC:*
MPI
Chad Brenner(MPI)
NIH-DHHS-US
09/2026 - 07/2031
$3,265,378

T23GM141746:*Biomedical Informatics and Data Science Training Program (BIDS-TP):*
MPI
Ivo Dinov(MPI)
NIH-DHHS-US
07/2026 - 06/2031
$2,703,355

R01NS145291:*Characterization and functional impact of somatic numtogenesis in the human cortex:*
MPI
Alan P. Boyle(MPI)
NIH-DHHS-US
06/2026 - 03/2031
$3,012,530

R03AG087485:*Leveraging large datasets to determine genomic impact of transposable elements in Alzheimers Disease and related dementias:*
Co-I (Principal Investigator:Weichen Zhou)
NIH-DHHS-US
06/2025 - 05/2027
$303,449

R01DE032699:*Defining the Role of HPV Integration Structures in HNSCC Molecular Heterogeneity:*
MPI
Chad Brenner(MPI)
NIH-DHHS-US
08/2024 - 04/2029
$2,931,889

UG3NS132084/UH3NS132084::*Molecular and Computational Tools for Identifying Somatic Mosaicism in Human Tissues:*
MPI
Alan P. Boyle(MPI)
NIH-DHHS-US
04/2023 - 04/2028
$2,713,053

U01HG011952:*Predicting the Impact of Genomic Variation on Cellular States:*
Co-I (Principal Investigator:Alan P. Boyle)
NIH-DHHS-US
08/2021 - 05/2027
$3,831,070

**Submitted - Open**

F31:*Evaluating Molecular Mechanisms and Clinical Utility of Circulating Tumor mRNA in HPV+ OPSCC:*
Consultant on
NIH-DHHS-US
12/2026 - 11/2029
$158,793

F31:*Mechanisms and Functional Impact of HPV-Host Fusion Genes in Oropharyngeal Cancer:*
Mentor
NIH-DHHS-US
12/2026 - 11/2027
$54,185

**Past Grants**

*Improving Bioinformatics Methods for Analysis of Virus-Associated Cancers:*
Co-I
Innovation in Cancer Informati
08/2022 - 08/2024
$224,000

N/A:*Repetitive elements in human health and disease:*
MPI
Taubman Institute Innovation Program (TIIP)
07/2022 - 06/2024
$659,764

T32GM141746:*Biomedical Informatics and Data Science Training Program (BIDS-TP):*
MPI
Ivo Dinov(MPI);Maureen Sartor(MPI)
NIH-DHHS-US
07/2021 - 06/2026
$2,096,667

R21HG011493:*New technologies for accurate capture and sequencing of repeat-associated regions:*
MPI
Alan P. Boyle(MPI)
NIH-DHHS-US
12/2020 - 11/2023
$643,024

F31DE030000:*Analysis of the E7-Mediated Mechanism of MHC Class I Repression in HNSCC:*
Consultant on
NIH-DHHS-US
07/2020 - 04/2023
$114,167

U24HG007497:*Identifying and Characterizing the Full Spectrum of Haplotype-resolved Structural Variation in Human Genome:*
PI of UM Site
NIH-DHHS-US-SubK sourced funding through The Jackson Laboratory
08/2019 - 05/2023
$597,831

F31HG010569:*Discovering Novel Structural Genomic Rearrangements Using Deep Neural Networks:*
Mentor
NIH-DHHS-US
04/2019 - 03/2022
$112,983

N/A:*Short Tandem repeats in precision health and human disease:*
MPI
Precision Health Investigators Award
01/2019 - 12/2020
$300,000

R35GM128836:*Mechanisms of translational control:*
Consultant on (Principal Investigator:Kristin Koutmou)
NIH-DHHS-US
08/2018 - 07/2024
$1,847,095

U41HG007497:*An Integrative Analysis of Structural Variation for the 1000 Genomes Project:*
PI of UM Site
NIH-DHHS-US-SubK sourced funding through The Jackson Laboratory
09/2017 - 08/2018
$84,451

R01AI118886:*Fidelity, robustness, and diversity in RNA virus evolution and pathogenesis:*
Co-I (Principal Investigator:Adam Lauring, MD, PhD)
NIH-DHHS-US
01/2016 - 12/2021
$2,023,562

U01MH106892:*2/3 Schizophrenia Genetics and Brain Somatic Mosaicism:*
Consultant on (Principal Investigator:John Moran)
NIH-DHHS-US
05/2015 - 01/2022
$4,706,350

F31NS090883:*Upstream open reading frames in neuronal function: a singular and genome-wide approach:*
Consultant on
NIH-DHHS-US
03/2015 - 02/2018
$102,399

U41HG007497:*An Integrative Analysis of Structural Variation for the 1000 Genomes Project:*
PI of UM Site
NIH-DHHS-US-SubK sourced funding through The Jackson Laboratory
09/2013 - 08/2018

$355,492

R01HG007068:*Discovery and Analysis of Structural Variation in Whole Genome Sequences:*
PI
NIH-DHHS-US
09/2013 - 07/2018
$1,509,405

R01GM103961:*Comprehensive Characterization of Canine Genomic Structural Diversity:*
Co-I (Principal Investigator:Jeffrey Kidd)
NIH-DHHS-US
09/2013 - 04/2017
$985,991

F32HG004207:*Improving INDEL Identification in Genomic Sequences:*
Funded by
National Human Genome Research Institute
09/2006 - 10/2008
$99,022

## Honors and Awards

**National**

| | |
|---|---|
| 2006 - 2008 | Ruth L. Kirschstein National Research Service Award (NRSA) Individual Postdoctoral Fellowship, Emory University |
| 2012 | Sixth Annual Young Investigators, GenomeWeb |
| 2015 | Highlighted in "Copy Number Analysis Starts to Add Up, Genetic Engineering & Biotechnology News |
| 2015 | Profiled, Journal of Young Investigators |
| 2018 | Profiled, Georgia Tech, College of Sciences |

**Institutional**

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|---|---|
| 2014 | Endowment for the Basic Sciences Teaching Award, Medical School |
| 2020 | Accelerator Award from the Endowment of Basic Sciences, Medical School |
| 2021 | Leadership Academy, Medical School, Office of Faculty Affairs & Faculty Development |

## Study Sections, Editorial Boards, Journal & Abstract Review

### Study Sections

**National**

| | |
|---|---|
| 2014 | NIH Study Section - Interpreting Variation in Human Non-Coding Genomic Regions Using Computational Approaches and Experimental Assessment, NIH, (Ad Hoc) |
| 2015 | NIH Study Section – Interpreting Variation in Human Non-Coding Genomic Regions Using Computational Approaches and Experimental Assessment, NIH, (Ad Hoc) |
| 2015 | NIH Study Section – Genomics, Computational Biology and Technology, NIH, (Ad Hoc) |

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|---|---|
| 2016 | NIH Study Section – Maximizing Investigators' Research Award for New and Early Stage Investigators, NIH, (Ad Hoc) |
| 2016 | NIH Study Section – Genomics, Computational Biology and Technology, NIH, (Ad Hoc) |
| 2018 | NIH Special Emphasis Panel - Rare Genetic Disorders as a Window into the Genetic Architecture of Mental Disorders (Co-Chair), NIH, (Ad Hoc) |
| 2019 | NIH Study Section – Genomics, Computational Biology and Technology (February), NIH, (Ad Hoc) |
| 2019 | NIH Study Section – Genomics, Computational Biology and Technology (October), NIH, (Ad Hoc) |
| 2022 | NIH Study Section - Bioengineering, Biodata, and Biomodeling Technologies, NIH, (Ad Hoc) |
| 2022 | Special Emphasis Panel - Expert-Driven Small Projects to Strengthen Gabriella Miller Kids First Discovery, NIH, (Ad Hoc) |
| 2024 | NHGRI Training Grant Review, NIH, (Ad Hoc) |
| 2025 | NIH Study Section - Bioengineering, Biodata, and Biomodeling Technologies (Co-Chair), NIH, (Ad Hoc) |
| 2026 | NIH Study Section - Bioengineering, Biodata, and Biomodeling Technologies (Chair), NIH, (Ad Hoc) |
| 2026 | NIH Study Section - Bioengineering, Biodata, and Biomodeling Technologies (Chair), NIH, (Ad Hoc) |
| 2026 | NIH Study Section - Biodata Management and Computational Modeling (Chair), NIH, (Ad Hoc) |

**Institutional**

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|---|---|
| 2018 | University of Michigan Precision Health Investigators Awards - Reviewer, Medical School, (Ad Hoc) |
| 2022 | PRR Trainee Committee - Application Review, University of Michigan, (Ad Hoc) |

## Editorial Boards / Journal & Abstract Reviews

**Editorial Boards**

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|---|---|
| 2015 - 2019 | Editorial Board Member, Scientific Reports |
| 2017 - 2020 | Editorial Board Member, PeerJ |

**Journal Review**

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|---|---|
| 2012 - Present | Bioinformatics (Ad Hoc) |
| 2012 - Present | BMC Bioinformatics (Ad Hoc) |
| 2012 - Present | Genome Biology (Ad Hoc) |
| 2012 - Present | Genome Research (Ad Hoc) |
| 2012 - Present | American Journal of Human Genetics (Ad Hoc) |
| 2012 - Present | Nature Genetics (Ad Hoc) |
| 2013 - Present | Nature Communications (Ad Hoc) |
| 2013 - Present | Nature Protocols (Ad Hoc) |
| 2014 - Present | Human Genetics (Ad Hoc) |
| 2015 - Present | PLoS Computational Biology (Ad Hoc) |
| 2016 - Present | Methods (Ad Hoc) |

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| 2018 - Present | European Journal of Human Genetics (Ad Hoc) |
| 2020 - Present | Genetics (Ad Hoc) |
| 2021 - Present | Nature (Ad Hoc) |

## Teaching

### Mentorship

**Faculty Member**

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|---|---|
| 01/2020 - Present | Weichen Zhou, University of Michigan, Computational Medicine and Bioinformatics, Research Investigator |

**Postdoctoral Fellow**

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|---|---|
| 01/2012 - 01/2016 | Gargi Dayama, University of Michigan, Computational Medicine and Bioinformatics, Postdoctoral Fellow |
| 01/2015 - 01/2020 | Weichen Zhou, University of Michigan, Computational Medicine and Bioinformatics, Postdoctoral Fellow |
| 01/2019 - 01/2020 | Yifan Wang, University of Michigan, Computational Medicine and Bioinformatics, Postdoctoral Fellow |

**Rotating Graduate Students**

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|---|---|
| 01/2014 | Fan Zhang, University of Michigan, Computational Medicine and Bioinformatics, PhD Rotation Student |
| 01/2017 | Catherine Barnier, University of Michigan, Computational Medicine and Bioinformatics, PhD Rotation Student |
| 01/2021 | Kai Li, University of Michigan, Computational Medicine and Bioinformatics, PhD Rotation Student |
| 01/2022 | Maya Bose, University of Michigan, Computational Medicine and Bioinformatics, PhD Rotation Student |
| 10/2025 - 12/2025 | Kobe Howcroft, University of Michigan, Computational Medicine and Bioinformatics, PIBS rotation student |
| 10/2025 - 12/2025 | Vital Nyabashi, University of Michigan, Computational Medicine and Bioinformatics, PIBS rotation student |

**Graduate Students – PhD**

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|---|---|
| 01/2013 - 01/2017 | Xuefang Zhao, University of Michigan, Computational Medicine and Bioinformatics, PhD Degree |
| 01/2013 - 01/2018 | Sang Chun, University of Michigan, Computational Medicine and Bioinformatics, PhD Degree |
| 01/2014 - 01/2018 | Akima George, University of Michigan, Computational Medicine and Bioinformatics, MS Degree |
| 01/2014 - 01/2019 | Yifan Wang, University of Michigan, Human Genetics, PhD Degree |
| 01/2015 - 01/2022 | Marcus Sherman, University of Michigan, Computational Medicine and Bioinformatics, PhD Degree |
| 01/2016 - 01/2022 | Alexandra Weber, University of Michigan, Computational Medicine and Bioinformatics, PhD Degree |
| 01/2017 - 01/2023 | Chen Sun, University of Michigan, Computational Medicine and Bioinformatics, PhD Student |
| 01/2019 - 04/2025 | Steve Ho, University of Michigan, Human Genetics, PhD Student |
| 01/2019 - 01/2024 | Wenjin Gu, University of Michigan, Computational Medicine and Bioinformatics, |

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|---|---|
| | PhD Student |
| 08/2022 - Present | Brandt Bessell, University of Michigan, Computational Medicine and Bioinformatics, PhD Student |
| 01/2023 - Present | Jinhao Wang, University of Michigan, Computational Medicine and Bioinformatics, PhD Degree |
| 06/2023 - Present | Xiaomeng Du, University of Michigan, Computational Medicine and Bioinformatics, PhD Student |
| 09/2025 - Present | Xinyi Deng, University of Michigan, Computational Medicine and Bioinformatics, Co-Mentor with Chad Brenner |

**Graduate Students – Master's**

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|---|---|
| 01/2016 - 01/2017 | Nan Lin, University of Michigan, Computational Medicine and Bioinformatics, MS Degree |
| 01/2017 - 05/2018 | Zhenning Zhang, University of Michigan, Computational Medicine and Bioinformatics, MS Degree |
| 01/2022 | Shaomiao Xia, University of Michigan, Computational Medicine and Bioinformatics, MS Student |
| 01/2022 - 11/2022 | Jinhao Wang, University of Michigan, Biostatistics, MS Student |
| 01/2026 - Present | Samantha Reimer, University of Michigan, Computational Medicine and Bioinformatics, MS Degree |

**Undergraduate Student**

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|---|---|
| 01/2018 - 01/2019 | Byungjun Kim, Cornell University, Summer Research |
| 01/2019 | Samantha Rondeau, University of Michigan, Research Experience |
| 01/2020 - 01/2021 | Priya Ghandi, University of Michigan, Research Experience |
| 01/2021 - 01/2022 | Yanming Gan, University of Michigan, Research Experience |
| 01/2022 | Thomas Chang, University of Michigan, Summer Research |
| 08/2022 - 05/2023 | Sophia Marcotte, University of Michigan, Research Experience |

## Teaching Activity

**National**

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|---|---|
| 06/2014 - 06/2019 | Lecturer, Mathematical and Theoretical Biology Institute, Arizona State University |

**Institutional**

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|---|---|
| 09/2012 | Lecturer, HG 632 – Experimental Genetics Systems, University of Michigan |
| 09/2012 - 12/2018 | Coursemaster/Lecturer, BIOL 527 – Introduction to Bioinformatics, University of Michigan |
| 01/2014 | Lecturer, UM NIEHS P30 Center and UM BRCF Bioinformatics Core Workshop – Introduction to Genome Variation, University of Michigan |
| 09/2014 | Lecturer, Coursera (online) – Introduction to Bioinformatics, University of Michigan |
| 01/2015 - 05/2019 | Lecturer, BIOINF 525 - Foundations in Bioinformatics and Systems Biology, University of Michigan |
| 08/2015 - 08/2019 | Coursemaster/Lecturer, BIOINF/HUMGEN/BIOSTATS 606 – Introduction to Biocomputing, University of Michigan |
| 03/2016 - Present | HUMGEN 803 – Current Methods, University of Michigan, Human Genetics, Lecturer |
| 03/2016 - Present | HUMGEN 551 – Computational Genomics, University of Michigan, Lecturer |

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| 01/2019 - Present | BIOINF 529 - Bioinformatics Concepts and Algorithms, University of Michigan, Coursemaster/Lecturer |
| 01/2021 - Present | PIBS 503 - Responsible Conduct in Research, University of Michigan, Session Leader |
| 02/2024 | IHPI Early Career Faculty Workshop, University of Michigan, Institute for Healthcare Policy & Innovation, Presenter |
| 09/2026 - Present | HUMGEN 821/822 - Student Seminar, University of Michigan, Human Genetics, Student Mentor |

## Dissertation Committees

**Preliminary Committee**

| | |
|---|---|
| 03/2012 | Yanxiao Zhang, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2012 | Brendan Veeneman, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2012 | Ellen Schmidt, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2013 | Andy Kong, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 09/2013 | Yindra Puentes, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 04/2014 | Shiya Song, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2014 | Ridvan Eksi, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2014 | Raymond Cavalcante, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 04/2015 | Laura Seaman, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 05/2015 | Fan Zhang, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2015 | Alexander Kalinin, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 05/2016 | Ricardo D'Oliveira Albanus, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 05/2016 | Jed Carlson, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 05/2016 | Sierra Nishizaki, University of Michigan, Human Genetics, Chair |
| 05/2016 | Vo Nguyen, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2016 | Shriya Sethuraman, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 12/2016 | Shengcheng Dong, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 04/2017 | Peter Orchard, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 05/2017 | Scott Ronquist, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |

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| 08/2017 | Ningxin Ouyang, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 09/2017 | Christopher Castro, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 05/2018 | Marlena Duda, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 05/2018 | Kevin Hu, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 11/2019 | Cathy Smith, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 12/2019 | Fan Feng, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 03/2020 | Ali Farhat, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 05/2020 | Michael Mumphrey, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 05/2020 | Renaid Kim, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 07/2020 | Chao Gao, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 08/2020 | Yuanhao Huang, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 12/2020 | Noshad Hosseini, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 07/2021 | Jialin Liu, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 08/2021 | Ford Hannum, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2021 | Shiting Li, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 04/2022 | Cooper Stansbury, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 05/2022 | Ahmed Elhossiny, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 07/2022 | Yueyang Shen, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2022 | Kinsey Van Deynze, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 08/2022 | Emily Koch, University of Michigan, Human Genetics, Committee Member |
| 04/2023 | Chinmay Raut, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 05/2023 | Bailey Garb, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 11/2023 | Katarina Pavlovic, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 01/2024 | Maria Lawas, University of Michigan, Computational Medicine and Bioinformatics, Chair |

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| 04/2024 | Rintsen Sherpa, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 05/2024 | Xiayan Li, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 05/2024 | Zhaowei Han, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 12/2024 | Steve Losh, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 12/2024 | Shaomiao Xia, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 04/2025 | Manasa Yadavalli, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 04/2025 | Ashley Bielawski, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 04/2025 | Ingrid Flaspohler, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 05/2026 | Ziyin Huang, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2026 | Junfeng Yang, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |

**Dissertation Committee**

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|---|---|
| 05/2013 - 04/2017 | Xuefang Zhao, Understanding the complexity of human structural genomic variation through multiple whole genome sequencing platforms, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 05/2013 - 11/2017 | Sang Chun, Development and Application of Next-Generation Sequencing Methods to Profile Cellular Translational Dynamics, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 09/2013 - 09/2016 | Brendan Veeneman, Development and application of methods to discover cancer-associated transcript variants, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 05/2014 - 05/2019 | Yifan Wang, Detection of Rare Events in Complex Sequencing Data, University of Michigan, Human Genetics, Chair |
| 07/2014 - 03/2016 | Kärt Tomberg, Identification of Thrombosis Modifier Genes Using ENU Mutagenesis in the Mouse, University of Michigan, Human Genetics, Committee Member |
| 08/2014 - 04/2016 | Killeen Kirkconnell, Capturing transcriptional dynamics using nascent RNA sequencing, University of Michigan, Human Genetics, Committee Member |
| 09/2014 - 03/2018 | Caitlin Rodriguez, The role of upstream open reading frames in regulating neuronal protein synthesis, University of Michigan, Neuroscience, Committee Member |
| 09/2014 - 09/2017 | Diane Flasch, LINE-1 Integration Preferences in Human Somatic Cells, University of Michigan, Human Genetics, Committee Member |
| 04/2015 - 04/2017 | Andy Kong, Computational strategies for proteogenomic analyses, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 10/2015 - 09/2017 | Daniel H. Hovelson, Precision oncology opportunities and disease insights from next-generation sequencing profiling of routine clinical biospecimens, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 01/2016 - 07/2022 | Marcus Sherman, Cultivation of enhanced bioinformatic-specific pedagogical |

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| | manipulatives, interventions, and professional development, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 06/2016 - 04/2019 | Fan Zhang, Leveraging Genetic Variants for Rapid, Robust, and Scalable Analysis of Massive Sequence Datasets, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2016 - 09/2020 | Sierra Nishizaki, Decoding the Non-coding Genome: Novel Technologies for the Characterization of Non-coding Elements and Variation, University of Michigan, Human Genetics, Committee Member |
| 09/2016 - 05/2022 | Alexandra Weber, Integrating Diverse Technologies for Genomic Variant Discovery, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 10/2016 - 05/2021 | Nguyen Vo, The Genetic Heterogeneity and Drug Resistance Mechanisms in Relapse Refractory Multiple Myeloma, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 10/2016 - 04/2018 | Adrian Tan, Statistical and Computational Methods for the Unified Analysis of Short Genetic Variants, University of Michigan, Biostatistics, Committee Member |
| 04/2017 - 03/2021 | Shengcheng Dong, Computational methods to identify regulatory variants in the non-coding regions of the human genome, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 05/2017 - 06/2021 | Hillary Miller, Regulation of stress-induced longevity, University of Michigan, Cellular and Molecular Biology, Committee Member |
| 01/2018 - 03/2023 | Chris Castro, Investigating the Role of Noncoding De Novo Single-Nucleotide Variants in Autism Spectrum Disorder, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 03/2018 - 06/2019 | Scott Ronquist, Methods for Analyzing the 4D Nucleome, with Application to Cellular Reprogramming, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 03/2018 - 04/2023 | Chen Sun, Development of computational methods for identifying mosaic copy number variation with single-cell sequencing, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 04/2018 - 01/2022 | Ningxin Ouyang, Deciphering Transcriptional Regulatory Circuits: Transcription Factor Binding and Regulatory Variants Identification, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 12/2018 - 05/2021 | Siyu Liu, Epigenetic effects in head and neck cancer and di-2-ethylhexyl phthalate (DEHP) exposure, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 01/2019 - 03/2025 | Steve Ho, Applied Machine Learning for Big Data Genomics, University of Michigan, Human Genetics, Chair |
| 01/2019 - 01/2024 | Wenjin Gu, Development of Viral Integration Analysis Technologies for Virus-Associated Cancer Research, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 01/2019 - 07/2020 | James Delorme, Linking intracellular events to network reorganization in sleep-dependent memory consolidation, University of Michigan, Neuroscience, Committee Member |
| 05/2019 - 06/2025 | Kevin Hu, Credentialing Genomic Fidelity in Mouse Cancer Models: Approaches and Applications, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2019 - 03/2023 | Elizabeth Gensterblum-Miller, Molecular Analysis of Recurrent Translocations in Mucoepidermoid Carcinoma, University of Michigan, Cellular and Molecular Biology, Committee Member |

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| 03/2020 - 07/2022 | Daniel Geislzer, Computational Methods for Characterizing Post-translational and Chemical Modifications Found in Open Searches, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 11/2020 - 03/2025 | Camille Mumm, Exploring Repetitive Elements in Neurodegenerative Disease using Targeted Long-Read Sequencing, University of Michigan, Human Genetics, Committee Member |
| 04/2021 - 01/2025 | Ali Farhat, Development and application of stochastic methods for modeling DNA-binding protein specificity and dynamics, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 06/2021 - Present | Christian Kelley, University of Michigan, Biophysics, Committee Member |
| 09/2021 - Present | Samantha Grudzien, University of Michigan, Neuroscience, Committee Member |
| 10/2021 - Present | Vinodh Balendran, University of Michigan, Neuroscience, Committee Member |
| 02/2022 - 06/2024 | D. Ford Hannum, Application of Emerging Transcriptomics Technologies and Methods to Gain Insight into the Human Reproductive System, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 03/2022 - 03/2025 | Shiting Li, Toward precision oncology for HPV-associated head and neck cancer (HNC): multi-omics analysis and machine learning, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 04/2022 - 02/2025 | Anthony Nguyen, An Investigation of Gene Duplications in Canines, University of Michigan, Human Genetics, Committee Member |
| 05/2022 - Present | Itzaira Mercado-Hernandez, University of Michigan, Human Genetics, Committee Member |
| 06/2022 - 12/2024 | Noshad Hosseini, Development of Novel Methods to Study Chromosomal Instability and Genetic Heterogeneity for Biomarker Identification, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 11/2022 - Present | Matthew Blacksmith, University of Michigan, Human Genetics, Committee Member |
| 12/2022 - Present | Jinhao Wang, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 04/2023 - Present | Brandt Bessell, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 04/2023 - Present | Xiaomeng Du, University of Michigan, Computational Medicine and Bioinformatics, Chair |
| 06/2023 - Present | Noah Helton, University of Michigan, Human Genetics, Committee Member |
| 10/2023 - Present | Chinmay Raut, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 07/2024 - Present | Lishi Yin, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 08/2024 - Present | Jianhui Gong, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 10/2024 - Present | Maria Lawas, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 10/2024 - Present | Rintsen Sherpa, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 10/2024 - Present | Monica Holmes, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 02/2025 - Present | Auden Bahr, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |

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| 10/2025 - Present | Shaomiao Xia, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 12/2025 - Present | Rosella Stower, University of Michigan, Biological Chemistry, Committee Member |
| 12/2025 - Present | Rosina Carr, University of Michigan, Computational Medicine and Bioinformatics, Committee Member |
| 03/2026 - Present | Dan Ciotlos, University of Michigan, Biostatistics, Committee Member |
| 07/2026 - Present | Jordan Currie, University of Michigan, Cellular and Molecular Biology, Committee Member |

## Memberships in Professional Societies

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|---|---|
| 2009 - Present | American Society of Human Genetics |
| 2011 - Present | International Society for Computational Biology |

## Committee/Service

**National**

| | |
|---|---|
| 2011 | Program Committee, First RECOMB Satellite Workshop on Massively Parallel Sequencing, Member |
| 2015 - Present | Program Committee, Great Lakes Bioinformatics Conference, Member |
| 2015 | Program Committee, Fifth RECOMB Satellite Workshop on Massively Parallel Sequencing, Member |
| 2016 | Abstract Committee, American Society of Human Genetics, Other, Abstract Reviewer |
| 2018 - 2023 | Program Committee, Intl. Conference on Algorithms for Computational Biology, Member |
| 2018 | Abstract Committee, American Society of Human Genetics, Other, Abstract Reviewer |
| 2019 - 2023 | National Association of Wabash Men, Wabash College, Board of Directors |

**Institutional**

| | |
|---|---|
| 2015 - 2016 | Faculty Leading Change, U-M ADVANCE Program, Other, Participant |
| 2017 - Present | Basic Research IT (BRIT) Committee, Medical School, Co-Chair |
| 2018 - 2020 | HITS Learning Services Governance Committee, Medical School, Member |
| 2019 - 2020 | PIBS Curriculum Committee, Medical School, Member |
| 2020 - 2022 | Advisory Committee on Appointments, Promotions, and Tenure (ACAPT), Medical School, Member |
| 2020 - 2022 | Pedagogy of Interdisciplinary Science Education (POISE) Training Program, University of Michigan, Advisory Board |
| 2022 | Grievance Hearing Board, University of Michigan, Member |
| 2022 - 2023 | Advisory Committee on Appointments, Promotions, and Tenure (ACAPT), Medical School, Chair |
| 2024 - 2025 | MM Gen AI Task Force, Education Work Group, University of Michigan, Co-Chair |
| 2026 - Present | MM CTO Search Committee, University of Michigan, Member |

**Departmental**

| | |
|---|---|
| 2012 | Master's Program Admissions Committee, DCM&B, Member |
| 2013 - 2020 | Master's Admission Committee, Human Genetics, Member |

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|---|---|
| 2014 - 2016 | Seminar Series Committee, DCM&B, Co-Chair |
| 2014 - 2017 | Picnic Committee, Human Genetics, Chair |
| 2014 - 2021 | PhD Admissions Committee, DCM&B, Member |
| 2016 - Present | Chair's Advisory Committee, DCM&B, Member |
| 2016 - 2019 | Website Committee, DCM&B, Chair |
| 2019 - 2026 | Space Committee, DCM&B, Member |
| 2021 - Present | Faculty Search Committee, DCM&B, Member |
| 2021 - Present | IT Committee, DCM&B, Chair |
| 2022 - 2023 | PhD Admissions Committee, DCM&B, Co-Chair |
| 2022 | Retreat Committee, DCM&B, Chair |

## Scholarly Activities

## Presentations

**Extramural Invited Presentation**

**Speaker**

1. Improving gene annotation of cytomegalovirus genomes by statistical and comparative genomics as verified by a proteomics-based analysis of isolated MCMV virions, **Mills R**, 9th International Cytomegalovirus Workshop, 05/2003, Maastricht, Netherlands

2. Designing Custom CGH Arrays: Considerations for CNV Discovery and Genotyping, **Mills R**, Agilent Technologies Workshop, American Society of Human Genetics, 10/2009, Honolulu, HI

3. Capturing Structural Variation from Whole Genome Population-Scale Sequencing: Perspectives from the 1000 Genomes Project, **Mills R**, Cambridge Healthtech Institute, X-GEN Congress and Expo, 03/2011, San Diego, CA

4. Challenges in Mapping Copy Number Variation from Population-scale Genome Sequence Data, **Mills R**, Open Science Grid, 2011 All Hands Meeting, 03/2011, Boston, MA

5. Structural Variation in the 1000 Genomes Project, **Mills R**, Genomic Disorders 2012, Wellcome Trust Sanger Institute, 03/2012, Hinxton, UK

6. Discovery and functional impact of structural variation across 1000 genomes, **Mills R**, Cambridge Healthtech Institute, NGx: Applying Next Generation Sequencing, 08/2012, Providence, RI

7. Discovery and Analysis of Structural Genomic Variation in Human Populations, **Mills R**, School of Biology Seminar, Wabash College, 10/2012, Crawfordsville, IN

8. 1000 Genomes Project Data Tutorial, **Mills R**, International Congress of Human Genetics, 11/2012, Montreal, Canada

9. 1000 Genomes Project Data Tutorial, **Mills R**, American Society of Human Genetics Annual Meeting, 11/2012, San Francisco, CA

10. Mapping structural variation by population-scale genome sequencing, **Mills R**, Radiation Effects Research Foundation, 03/2013, Hiroshima, Japan

11. Exploring Complex Structural Genomic Variation using Next-Gen Sequencing, **Mills R**, BioConference Live, Genetics and Genomics, 08/2014, Online (Live Virtual Presentation)

12. Genomic landscape of polymorphic nuclear mitochondrial insertions in humans and other primates, **Mills R**, American Society of Human Genetics Annual Meeting, 10/2014, San Diego, CA

13. Exploring the Hidden Genome: Deciphering Cryptic and Complex Structural Variation, **Mills R**, Ewha Womans University, 07/2016, Seoul, South Korea

14. Excavating the Deep Genome: Deciphering Structural Variation in Complex and Repetitive Regions,

**Mills R**, Oakland University, 04/2019, Rochester, MI

15. Identification and Characterization of Cryptic Structural Variation in Human Genomes, **Mills R**, Association for Molecular Pathology Annual Meeting & Expo, 11/2019, Baltimore, MD

16. Identification and Characterization of Cryptic Genomic Repetitive Elements, **Mills R**, Stanford University, 05/2020, Stanford, CA (Live Virtual Presentation)

17. Identification and Characterization of Structural Variation in Human Genomes, **Mills R**, Future Biotech Winter Retreat, 11/2020, Novosibirsk, Russia

18. New Technologies for Identifying and Characterizing Genomic Repetitive Elements, **Mills R**, Department of Biomedical Informatics Colloquium, University of Pittsburgh, 02/2021, Pittsburgh, PA (Live Virtual Presentation)

19. New Technologies for Identifying and Characterizing Genomic Repetitive Elements, **Mills R**, SV Working Group, Broad Institute, 02/2022, Boston, MA (Live Virtual Presentation)

20. Pebbles in the Sand: Exploring the Mosaic Nature of Neuronal Genomes, **Mills R**, CSL Student Conference, University of Illinois at Urbana-Champaign, 02/2023, Urbana-Champaign, IL

**Moderator**

1. Platform Moderator, **Mills R**, American Society of Human Genetics Annual Meeting, 11/2012, San Francisco, CA

2. Invited Session Moderator, **Mills R**, American Society of Human Genetics Annual Meeting, 10/2014, San Diego, CA

**Panel**

1. Panel Discussion: Copy Number Variants, **Mills R**, 3rd Annual PQG Conference, Harvard School of Public Health, 11/2009, Boston, MA

**Intramural Invited Presentation**

**Speaker**

1. Analysis of Structural Variationin the 1000 Genomes Project Pilot:New Methods, New Insights, **Mills R**, McCarroll S, Handsaker B, Medical and Population Genetics Seminar, Broad Institute, 09/2010, Boston, MA

2. Natural Structural Variation in the Human Genome, **Mills R**, Interdisciplinary Group Seminar (IGS), Rackham Graduate School, University of Michigan, 04/2012, Ann Arbor, MI

3. Navigating Genomic Complexity: Discovery and Analysis of Structural Variation, **Mills R**, NCIBI Tools and Technology Series, University of Michigan, 05/2012, Ann Arbor, MI

## Publications/Scholarship

(Co-First Author *; Corresponding author **; Co-Last author ***)

## Peer Reviewed

**Journal Article**

1. Borodovsky M, **Mills R**, Besemer J, Lomsadze A: Prokaryotic gene prediction using GeneMark and GeneMark.hmm. *Curr Protoc Bioinformatics*.Chapter 4: Unit4.5, 05/2003. PM18428700

2. Borodovsky M, Lomsadze A, Ivanov N, **Mills R**: Eukaryotic gene prediction using GeneMark.hmm. *Curr Protoc Bioinformatics*.Chapter 4: Unit4.6, 05/2003. PM18428701

3. Perelygina L, Zhu L, Zurkuhlen H, **Mills R**, Borodovsky M, Hilliard JK: Complete sequence and comparative analysis of the genome of herpes B virus (Cercopithecine herpesvirus 1) from a rhesus monkey. *Journal of Virology*.77(11): 6167-6177, 06/2003. PM12743273

4. **Mills R**, Rozanov M, Lomsadze A, Tatusova T, Borodovsky M: Improving gene annotation of complete viral genomes. *Nucleic Acids Res*.31(23): 7041-7055, 12/2003. PM14627837

5. Kattenhorn LM, **Mills R**, Wagner M, Lomsadze A, Makeev V, Borodovsky M, Ploegh HL, Kessler BM:

Identification of proteins associated with murine cytomegalovirus virions. *J Virol*.78(20): 11187-11197, 10/2004. PM15452238

6. **Mills RE**, Bennett EA, Iskow RC, Luttig CT, Tsui C, Pittard WS, Devine SE: Recently mobilized transposons in the human and chimpanzee genomes. *Am J Hum Genet*.78(4): 671-679, 04/2006. PM16532396

7. **Mills RE**, Luttig CT, Larkins CE, Beauchamp A, Tsui C, Pittard WS, Devine SE: An initial map of insertion and deletion (INDEL) variation in the human genome. *Genome Res*.16(9): 1182-1190, 09/2006. PM16902084

8. Lange A, **Mills RE**, Lange CJ, Stewart M, Devine SE, Corbett AH: Classical nuclear localization signals: definition, function, and interaction with importin alpha. *J Biol Chem*.282(8): 5101-5105, 02/2007. PM17170104

9. **Mills RE**, Bennett EA, Iskow RC, Devine SE: Which transposable elements are active in the human genome?. *Trends Genet*.23(4): 183-191, 04/2007. PM17331616

10. Lange A, **Mills RE**, Devine SE, Corbett AH: A PY-NLS nuclear targeting signal is required for nuclear localization and function of the Saccharomyces cerevisiae mRNA-binding protein Hrp1. *J Biol Chem*.283(19): 12926-12934, 05/2008. PM18343812

11. Bennett EA, Keller H, **Mills RE**, Schmidt S, Moran JV, Weichenrieder O, Devine SE: Active Alu retrotransposons in the human genome. *Genome Res*.18(12): 1875-1883, 12/2008. PM18836035

12. Kim J-I, Ju YS, Park H, Kim S, Lee S, Yi J-H, Mudge J, Miller NA, Hong D, Bell CJ, Kim H-S, Chung I-S, Lee W-C, Lee J-S, Seo S-H, Yun J-Y, Woo HN, Lee H, Suh D, Lee S, Kim H-J, Yavartanoo M, Kwak M, Zheng Y, Lee MK, Park H, Kim JY, Gokcumen O, **Mills RE**, Zaranek AW, Thakuria J, Wu X, Kim RW, Huntley JJ, Luo S, Schroth GP, Wu TD, Kim H, Yang K-S, Park W-Y, Kim H, Church GM, Lee C, Kingsmore SF, Seo J-S: A highly annotated whole-genome sequence of a Korean individual. *Nature*.460(7258): 1011-1015, 08/2009. PM19587683

13. Lange A, McLane LM, **Mills RE**, Devine SE, Corbett AH: Expanding the definition of the classical bipartite nuclear localization signal. *Traffic*.11(3): 311-323, 03/2010. PM20028483

14. Park H, Kim J-I, Ju YS, Gokcumen O, **Mills RE**, Kim S, Lee S, Suh D, Hong D, Kang HP, Yoo YJ, Shin J-Y, Kim H-J, Yavartanoo M, Chang YW, Ha J-S, Chong W, Hwang G-R, Darvishi K, Kim H, Yang SJ, Yang K-S, Kim H, Hurles ME, Scherer SW, Carter NP, Tyler-Smith C, Lee C, Seo J-S: Discovery of common Asian copy number variants using integrated high-resolution array CGH and massively parallel DNA sequencing. *Nat Genet*.42(5): 400-405, 05/2010. PM20364138

15. Iskow RC, McCabe MT, **Mills RE**, Torene S, Pittard WS, Neuwald AF, Van Meir EG, Vertino PM, Devine SE: Natural mutagenesis of human genomes by endogenous retrotransposons. *Cell*.141(7): 1253-1261, 06/2010. PM20603005

16. Mullaney JM, **Mills RE**, Pittard WS, Devine SE: Small insertions and deletions (INDELs) in human genomes. *Hum Mol Genet*.19(R2): R131-R136, 10/2010. PM20858594

17. **1000 Genomes Project Consortium**, Abecasis GR, Altshuler D, Auton A, Brooks LD, Durbin RM, Gibbs RA, Hurles ME, McVean GA: A map of human genome variation from population-scale sequencing. *Nature*.467(7319): 1061-1073, 10/2010. PM20981092

18. Sudmant PH, Kitzman JO, Antonacci F, Alkan C, Malig M, Tsalenko A, Sampas N, Bruhn L, Shendure J, **1000 Genomes Project**, Eichler EE: Diversity of human copy number variation and multicopy genes. *Science*.330(6004): 641-646, 10/2010. PM21030649

19. Gokcumen O, Babb PL, Iskow RC, Zhu Q, Shi X, **Mills RE**, Ionita-Laza I, Vallender EJ, Clark AG, Johnson WE, Lee C: Refinement of primate copy number variation hotspots identifies candidate genomic regions evolving under positive selection. *Genome Biol*.12(5): R52, 01/2011. PM21627829

20. **Mills RE**, Pittard WS, Mullaney JM, Farooq U, Creasy TH, Mahurkar AA, Kemeza DM, Strassler DS, Ponting CP, Webber C, Devine SE: Natural genetic variation caused by small insertions and deletions in the human genome. *Genome Res*.21(6): 830-839, 06/2011. PM21460062

21. Pinto D, Darvishi K, Shi X, Rajan D, Rigler D, Fitzgerald T, Lionel AC, Thiruvahindrapuram B, MacDonald JR, **Mills R**, Prasad A, Noonan K, Gribble S, Prigmore E, Donahoe PK, Smith RS, Park JH,

Hurles ME, Carter NP, Lee C, Scherer SW, Feuk L: Comprehensive assessment of array-based platforms and calling algorithms for detection of copy number variants. *Nature Biotechnology*.29(6): 512-520, 06/2011. PM21552272

22. Conrad DF, Keebler JE M, DePristo MA, Lindsay SJ, Zhang Y, Casals F, Idaghdour Y, Hartl CL, Torroja C, Garimella KV, Zilversmit M, Cartwright R, Rouleau GA, Daly M, Stone EA, Hurles ME, Awadalla P, **1000 Genomes Project**: Variation in genome-wide mutation rates within and between human families. *Nat Genet*.43(7): 712-714, 06/2011. PM21666693

23. Gravel S, Henn BM, Gutenkunst RN, Indap AR, Marth GT, Clark AG, Yu F, Gibbs RA, **1000 Genomes Project**, Bustamante CD: Demographic history and rare allele sharing among human populations. *Proc Natl Acad Sci U S A*.108(29): 11983-11988, 07/2011. PM21730125

24. Marth GT, Yu F, Indap AR, Garimella K, Gravel S, Leong WF, Tyler-Smith C, Bainbridge M, Blackwell T, Zheng-Bradley X, Chen Y, Challis D, Clarke L, Ball EV, Cibulskis K, Cooper DN, Fulton B, Hartl C, Koboldt D, Muzny D, Smith R, Sougnez C, Stewart C, Ward A, Yu J, Xue Y, Altshuler D, Bustamante CD, Clark AG, Daly M, DePristo M, Flicek P, Gabriel S, Mardis E, Palotie A, Gibbs R, **1000 Genomes Project**: The functional spectrum of low-frequency coding variation. *Genome Biol*.12(9): R84, 09/2011. PM21917140

25. Chen X, Shi X, Xu X, Wang Z, **Mills R**, Lee C, Xu J: A two-graph guided multi-task Lasso approach for eQTL mapping. *Proceedings of the 15th International Conference of Artificial Intelligence and Statistics (AISTATS)*.22: 208-217, 01/2012

26. Brown KH, Dobrinski KP, Lee AS, Gokcumen O, **Mills RE**, Shi X, Chong WW S, Chen JY H, Yoo P, David S, Peterson SM, Raj T, Choy KW, Stranger BE, Williamson RE, Zon LI, Freeman JL, Lee C: Extensive genetic diversity and substructuring among zebrafish strains revealed through copy number variant analysis. *Proc Natl Acad Sci U S A*.109(2): 529-534, 01/2012. PM22203992

27. MacArthur DG, Balasubramanian S, Frankish A, Huang N, Morris J, Walter K, Jostins L, Habegger L, Pickrell JK, Montgomery SB, Albers CA, Zhang ZD, Conrad DF, Lunter G, Zheng H, Ayub Q, DePristo MA, Banks E, Hu M, Handsaker RE, Rosenfeld JA, Fromer M, Jin M, Mu XJ, Khurana E, Ye K, Kay M, Saunders GI, Suner M-M, Hunt T, Barnes IH A, Amid C, Carvalho-Silva DR, Bignell AH, Snow C, Yngvadottir B, Bumpstead S, Cooper DN, Xue Y, Romero IG, **1000 Genomes Project Consortium**, Wang J, Li Y, Gibbs RA, McCarroll SA, Dermitzakis ET, Pritchard JK, Barrett JC, Harrow J, Hurles ME, Gerstein MB, Tyler-Smith C: A systematic survey of loss-of-function variants in human protein-coding genes. *Science*.335(6070): 823-828, 02/2012. PM22344438

28. Chiang C, Jacobsen JC, Ernst C, Hanscom C, Heilbut A, Blumenthal I, **Mills RE**, Kirby A, Lindgren AM, Rudiger SR, McLaughlan CJ, Bawden CS, Reid SJ, Faull RL M, Snell RG, Hall IM, Shen Y, Ohsumi TK, Borowsky ML, Daly MJ, Lee C, Morton CC, MacDonald ME, Gusella JF, Talkowski ME: Complex reorganization and predominant non-homologous repair following chromosomal breakage in karyotypically balanced germline rearrangements and transgenic integration. *Nat Genet*.44(4): 390-S1, 03/2012. PM22388000

29. Clarke L, Zheng-Bradley X, Smith R, Kulesha E, Xiao C, Toneva I, Vaughan B, Preuss D, Leinonen R, Shumway M, Sherry S, Flicek P, **1000 Genomes Project Consortium**: The 1000 Genomes Project: data management and community access. *Nat Methods*.9(5): 459-462, 04/2012. PM22543379

30. Iskow RC, Gokcumen O, Abyzov A, Malukiewicz J, Zhu Q, Sukumar AT, Pai AA, **Mills RE**, Habegger L, Cusanovich DA, Rubel MA, Perry GH, Gerstein M, Stone AC, Gilad Y, Lee C: Regulatory element copy number differences shape primate expression profiles. *Proc Natl Acad Sci U S A*.109(31): 12656-12661, 07/2012. PM22797897

31. **1000 Genomes Project Consortium**, Abecasis GR, Auton A, Brooks LD, DePristo MA, Durbin RM, Handsaker RE, Kang HM, Marth GT, McVean GA: An integrated map of genetic variation from 1,092 human genomes. *Nature*.491(7422): 56-65, 11/2012. PM23128226

32. Xue Y, Chen Y, Ayub Q, Huang N, Ball EV, Mort M, Phillips AD, Shaw K, Stenson PD, Cooper DN, Tyler-Smith C**, **Consortium T G P**: Deleterious- and Disease-Allele Prevalence in Healthy Individuals: Insights from Current Predictions, Mutation Databases, and Population-Scale Resequencing. *American Journal of Human Genetics*.91(6): 1022-1032, 12/2012. PM23217326

33. Montgomery SB, Goode DL, Kvikstad E, Albers CA, Zhang ZD, Mu XJ, Ananda G, Howie B, Karczewski KJ, Smith KS, Anaya V, Richardson R, Davis J, **1000 GP C**, MacArthur DG, Sidow A, Duret L, Gerstein M, Makova KD, Marchini J, McVean G, Lunter G: The origin, evolution, and functional impact of short insertion-deletion variants identified in 179 human genomes. *Genome research*.23(5)01/2013. PM23478400

34. Rogers AJ, Chu J-H, Darvishi K, Ionita-Laza I, Lehmann H, **Mills R**, Lee C, Raby BA: Copy number variation prevalence in known asthma genes and their impact on asthma susceptibility. *Clin Exp Allergy*.43(4): 455-462, 04/2013. PM23517041

35. Chu J-H, Rogers A, Ionita-Laza I, Darvishi K, **Mills RE**, Lee C, Raby BA: Copy number variation genotyping using family information. *BMC Bioinformatics*.14: 157, 05/2013. PM23656838

36. Todd PK**, Oh SY, Krans A, He F, Sellier C, Frazer M, Renoux AJ, Chen K-C, Scaglione KM, Basrur V, Elenitoba-Johnson K, Vonsattel JP, Louis ED, Sutton MA, Taylor JP, **Mills RE**, Charlet-Berguerand N, Paulson HL**: CGG Repeat-Associated Translation Mediates Neurodegeneration in Fragile X Tremor Ataxia Syndrome. *Neuron*.79(2): 402-402, 07/2013. DOI:10.1016/j.neuron.2013.07.008

37. Gokcumen O, Tischler V, Tica J, Zhu Q, Iskow RC, Lee E, Fritz MH-Y, Langdon A, Stütz AM, Pavlidis P, Benes V, **Mills RE**, Park PJ, Lee C, Korbel JO: Primate genome architecture influences structural variation mechanisms and functional consequences. *Proc Natl Acad Sci U S A*.110(39): 15764-15769, 09/2013. PM24014587

38. Khurana E, Fu Y, Colonna V, Mu XJ, Kang HM, Lappalainen T, Sboner A, Lochovsky L, Chen J, Harmanci A, Das J, Abyzov A, Balasubramanian S, Beal K, Chakravarty D, Challis D, Chen Y, Clarke D, Clarke L, Cunningham F, Evani US, Flicek P, Fragoza R, Garrison E, Gibbs R, Gümüş ZH, Herrero J, Kitabayashi N, Kong Y, Lage K, Liluashvili V, Lipkin SM, MacArthur DG, Marth G, Muzny D, Pers TH, Ritchie GR S, Rosenfeld JA, Sisu C, Wei X, Wilson M, Xue Y, Yu F, **1000 Genomes Project Consortium**, Dermitzakis ET, Yu H, Rubin MA, Tyler-Smith C, Gerstein M: Integrative annotation of variants from 1092 humans: application to cancer genomics. *Science*.342(6154): 1235587, 10/2013. PM24092746

39. Abyzov A, Iskow R, Gokcumen O, Radke DW, Balasubramanian S, Pei B, Habegger L, **1000 Genomes Project Consortium**, Lee C, Gerstein M: Analysis of variable retroduplications in human populations suggests coupling of retrotransposition to cell division. *Genome Res*.23(12): 2042-2052, 12/2013. PM24026178

40. Park H, Kim D, Kim CH, **Mills RE**, Chang MY, Iskow RC, Ko S, Moon JI, Choi HW, Yoo PS M, Do JT, Han MJ, Lee EG, Jung JK, Zhang C, Lanza R, Kim KS: Increased genomic integrity of an improved protein-based mouse induced pluripotent stem cell method compared with current viral-induced strategies. *Stem Cells Translational Medicine*.3(5): 599-609, 01/2014. PM24763686

41. Silva AG, Krepischi AC V, Torrezan GT, Capelli LP, Carraro DM, D'Angelo CS, Koiffmann CP, Zatz M, Naslavsky MS, Masotti C, Otto PA, Achatz MI W, **Mills RE**, Lee C, Pearson PL, Rosenberg C: Does germ-line deletion of the PIP gene constitute a widespread risk for cancer?. *European Journal of Human Genetics*.22(3): 307-309, 03/2014. PM23778870

42. Delaneau O, Marchini J, **1000 Genomes Project Consortium**, 1000 Genomes Project Consortium: Integrating sequence and array data to create an improved 1000 Genomes Project haplotype reference panel. *Nat Commun*.5: 3934, 06/2014. PM25653097

43. Colonna V, Ayub Q, Chen Y, Pagani L, Luisi P, Pybus M, Garrison E, Xue Y, Tyler-Smith C, **1000 Genomes Project Consortium**, Abecasis GR, Auton A, Brooks LD, DePristo MA, Durbin RM, Handsaker RE, Kang HM, Marth GT, McVean GA: Human genomic regions with exceptionally high levels of population differentiation identified from 911 whole-genome sequences. *Genome Biol*.15(6): R88, 06/2014. PM24980144

44. Brand H, Pillalamarri V, Collins RL, Eggert S, O'Dushlaine C, Braaten EB, Stone MR, Chambert K, Doty ND, Hanscom C, Rosenfeld JA, Ditmars H, Blais J, **Mills R**, Lee C, Gusella JF, McCarroll S, Smoller JW, Talkowski ME, Doyle AE: Cryptic and complex chromosomal aberrations in early-onset neuropsychiatric disorders. *Am J Hum Genet*.95(4): 454-461, 10/2014. PM25279985

45. Dayama G, Emery SB, Kidd JM, **Mills RE**: The genomic landscape of polymorphic human nuclear mitochondrial insertions. *Nucleic Acids Res*.42(20): 12640-12649, 11/2014. PM25348406

46. Jorge DM D M, **Mills RE**, Lauring AS: CodonShuffle: a tool for generating and analyzing synonymously mutated sequences. *Virus Evol*.1(1): vev012, 01/2015. PM27774284

47. Sudmant PH, Rausch T, Gardner EJ, Handsaker RE, Abyzov A, Huddleston J, Zhang Y, Ye K, Jun G, Fritz MH-Y, Konkel MK, Malhotra A, Stütz AM, Shi X, Casale FP, Chen J, Hormozdiari F, Dayama G, Chen K, Malig M, Chaisson MJ P, Walter K, Meiers S, Kashin S, Garrison E, Auton A, Lam HY K, Mu XJ, Alkan C, Antaki D, Bae T, Cerveira E, Chines P, Chong Z, Clarke L, Dal E, Ding L, Emery S, Fan X, Gujral M, Kahveci F, Kidd JM, Kong Y, Lameijer E-W, McCarthy S, Flicek P, Gibbs RA, Marth G, Mason CE, Menelaou A, Muzny DM, Nelson BJ, Noor A, Parrish NF, Pendleton M, Quitadamo A, Raeder B, Schadt EE, Romanovitch M, Schlattl A, Sebra R, Shabalin AA, Untergasser A, Walker JA, Wang M, Yu F, Zhang C, Zhang J, Zheng-Bradley X, Zhou W, Zichner T, Sebat J, Batzer MA, McCarroll SA, 1000 Genomes Project Consortium, **Mills RE**, Gerstein MB, Bashir A, Stegle O, Devine SE, Lee C, Eichler EE, Korbel JO: An integrated map of structural variation in 2,504 human genomes. *Nature*.526(7571): 75-81, 10/2015. PM26432246

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**Addendum**

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## Non-Peer Reviewed

**Comparative Study**

1. Sulovari A, Li R, Audano PA, Porubsky D, Vollger MR, Logsdon GA, Human Genome Structural Variation Consortium, Warren WC, Pollen AA, Chaisson MJ P, Eichler EE: Human-specific tandem repeat expansion and differential gene expression during primate evolution. *Proc Natl Acad Sci U S A*.116(46): 23243-23253, 11/2019. PM31659027

**Published Erratum**

1. Zook JM**, Hansen NF, Olson ND, Chapman L, Mullikin JC, Xiao C, Sherry S, Koren S, Phillippy AM, Boutros PC, Sahraeian SM E, Huang V, Rouette A, Alexander N, Mason CE, Hajirasouliha I, Ricketts C, Lee J, Tearle R, Fiddes IT, Barrio AM, Wala J, Carroll A, Ghaffari N, Rodriguez OL, Bashir A, Jackman S, Farrell JJ, Wenger AM, Alkan C, Soylev A, Schatz MC, Garg S, Church G, Marschall T, Chen K, Fan X, English AC, Rosenfeld JA, Zhou W, **Mills RE**, Sage JM, Davis JR, Kaiser MD, Oliver JS, Catalano AP, Chaisson MJ P, Spies N, Sedlazeck FJ, Salit M: Author Correction: A robust benchmark for detection of germline large deletions and insertions. *Nature Biotechnology*.38(11): 1357-1357, 11/2020. PM32699374

2. Rodin RE, Dou Y, Kwon M, Sherman MA, D'Gama AM, Doan RN, Rento LM, Girskis KM, Bohrson CL, Kim SN, Nadig A, Luquette LJ, Gulhan DC, Park PJ**, Walsh CA**: Author Correction: The landscape of somatic mutation in cerebral cortex of autistic and neurotypical individuals revealed by ultra-deep whole-genome sequencing. *Nature Neuroscience*.24(4): 611-611, 04/2021. PM33753946

3. Zhu X, Zhou B, Pattni R, Gleason K, Tan C, Kalinowski A, Sloan S, Fiston-Lavier A-S, Mariani J, Petrov D, Barres BA, Duncan L, Abyzov A, Vogel H, Moran JV, Vaccarino FM, Tamminga CA, Levinson DF, Urban AE**: Author Correction: Machine learning reveals bilateral distribution of somatic L1 insertions in human neurons and glia. *Nature Neuroscience*.26(10): 1833-1833, 10/2023. PM37648813

4. Logsdon GA, Ebert P, Audano PA, Loftus M, Porubsky D, Ebler J, Yilmaz F, Hallast P, Prodanov T, Yoo D, Paisie CA, Harvey WT, Zhao X, Martino GV, Henglin M, Munson KM, Rabbani K, Chin C-S, Gu B, Ashraf H, Scholz S, Austine-Orimoloye O, Balachandran P, Bonder MJ, Cheng H, Chong Z, Crabtree J, Gerstein M, Guethlein LA, Hasenfeld P, Hickey G, Hoekzema K, Hunt SE, Jensen M, Jiang Y, Koren S, Kwon Y, Li C, Li H, Li J, Norman PJ, Oshima KK, Paten B, Phillippy AM, Pollock NR, Rausch T, Rautiainen M, Song Y, Söylev A, Sulovari A, Surapaneni L, Tsapalou V, Zhou W, Zhou Y, Zhu Q, Zody MC, **Mills RE**, Devine SE, Shi X, Talkowski ME, Chaisson MJ P, Dilthey AT, Konkel MK, Korbel JO, Lee C, Beck CR, Eichler EE, Marschall T: Author Correction: Complex genetic variation in nearly complete human genomes. *Nature*.645(8081): E6, 09/2025. PM40858940

## Preprints

1. Ho SS**, **Mills RE****: Domain-specific embeddings uncover latent genetics knowledge, bioRxiv, PM40166296

2. Ho SS**, **Mills RE****: Contextualizing gene expression with feature rich graph neural networks, bioRxiv, DOI:10.1101/2025.03.17.643816

3. Zhou W, Mumm C, Gan Y, Switzenberg JA, Wang J, De Oliveira P, Kathuria K, Losh SJ, McDonald TL, Bessell B, Van Deynze K, McConnell MJ**, Boyle AP**, **Mills RE****: A personalized multi-platform assessment of somatic mosaicism in the human frontal cortex, DOI:10.1101/2024.12.18.629274

4. Gu W, Bhangale AD, Du X, Deng X, Gensterblum-Miller EU, Currie J, Brummel CV, Nyabashi V, Buchakjian M, Casper KA, Chinn SE, Forner D, Malloy KM, Mierzwa ML, Prince ME, Shah JL, Shuman AG, Stucken CL, Swiecicki PL, Worden FP, Yalamanchi P, McHugh JB, Kuhs KA, Boyle AP, Jiang H, Neal ME, Spector ME, **Mills RE****, Brenner CJ**: Spatiotemporal Heterogeneity of HPV Integration Architectures Drive Oncogenic Features of Oropharyngeal Cancer, bioRxiv, DOI:10.64898/2026.10.03.756294

5. Wang S, Bae M, Wang J, Zhao B, Nguyen K, Mallett S, Switzenberg JA, Losh SJ, Sexton CE, Miao B, Dong S, Zeng X, Wang Z, McDonald TL, Mumm C, Gadde RK, Tariq AM, Chen Z, Feng WC, Burn A, Park J, Chu C, Shen H, Wang T, Urban AE, Zhu X, Li H, Burns KH, Chun H-JE, Park PJ, SMaHT MEI Working Group, Boyle AP**, **Mills RE****, Zhou W**, Lee EA**: Multi-platform framework for mapping somatic retrotransposition in human tissues, 10/2025. PM41278868 (accepted for publication)

6. Somatic Mosaicism across Human Tissues Network, Abyzov A: Comprehensive benchmarking of somatic mutation detection by the SMaHT Network, https://www.biorxiv.org/, 10/2025. PM41279200 (accepted for publication)